

Weather-Conditioned Tropical Outage Prediction

Development Experiment Report

September 27, 2026

Status. Completed development evaluation; not an independent confirmation or an operational weather-forecast test.

1 Forecasting Results and Interpretation

The complete campaign comprises 70 neural fits and 75 tree fits. Each of the 1,633 county–event pairs is evaluated exactly once in an outer event fold. All results below use the fixed public development panel; the sealed confirmation outcomes were not accessed.

Table 1: Full 144-hour trajectory accuracy: design-weighted pooled metrics.

Model	MAE	RMSE
Zero forecast	0.00978	0.05805
Persistence	0.00973	0.05728
Damped persistence	0.00971	0.05729
Matched AsymODE host	0.01078	0.04858
Joint GCRK-rate model	0.01171	0.05073
ERA5 residual tree	0.01439	0.05345
Dual-weather residual tree	0.01197	0.04813
Dual-weather MAE tree	0.00954	0.04921
Inner-selected ensemble	0.01108	0.04584

The inner-selected ensemble changes pooled RMSE relative to the matched host by -5.65% (negative is better), with a paired family-bootstrap 95% interval [-17.02%, +4.24%]. It improves RMSE on 10 of 15 systems. The corresponding MAE change is +2.81%.

The joint GCRK-rate predictor changes RMSE relative to the matched host by +4.41%, with a 95% interval [+1.18%, +9.46%]. Because this arm changes the kernel and rate heads together, this is not an isolated test of geographic information.

The ensemble comparison does not establish a reliably positive gain at the stated interval level. Any favorable point estimate must therefore be interpreted as exploratory.

Against zero prediction, the ensemble reduces pooled RMSE by 21.03% (95% interval [10.85%, 31.29%]), but improves only 6 of 15 systems. Its RMSE reduction over the dual-weather residual tree is 4.75% (interval [-2.11%, 12.53%]).

The lowest full-path MAE belongs to the dual-weather absolute-loss tree (0.00954); its MAE reduction relative to the host is 11.51% (interval [3.57%, 27.68%]). Thus the RMSE-oriented ensemble and the absolute-loss tree represent different error tradeoffs. The present joint GCRK-rate arm does not support replacing the matched host.

Table 2: Development event-held-out accuracy on 15 tropical systems. Design-weighted pooled MAE and RMSE of the public outage fraction; lower is better. Weather inputs are conditional hindcast information.

Model	MAE				RMSE			
	+1 h	+6 h	+24 h	+48 h	+1 h	+6 h	+24 h	+48 h
Zero forecast	0.00088	0.00093	0.00993	0.01524	0.01588	0.01641	0.05660	0.07640
Persistence	0.00016	0.00029	0.00949	0.01498	0.00114	0.00187	0.05460	0.07539
Damped persistence	0.00016	0.00029	0.00948	0.01497	0.00114	0.00187	0.05460	0.07540
Matched AsymODE host	0.00053	0.00146	0.00921	0.01559	0.00512	0.01447	0.04026	0.05885
Joint GCRK-rate model	0.00051	0.00161	0.00969	0.01704	0.00415	0.01355	0.04216	0.06407
ERA5 residual tree	0.00145	0.00156	0.01025	0.02443	0.00686	0.00691	0.04490	0.07486
Dual-weather residual tree	0.00099	0.00105	0.00936	0.01924	0.00436	0.00455	0.04182	0.06625
Dual-weather MAE tree	0.00165	0.00169	0.00802	0.01514	0.01779	0.01816	0.04190	0.06249
Inner-selected ensemble	0.00099	0.00136	0.00860	0.01761	0.00573	0.01019	0.03744	0.06135

The four horizons are point predictions from the same 144-hour path; weights do not vary by horizon. The full-path RMSE ordering need not hold at every lead time. Persistence is particularly strong at the earliest leads in this panel. All seed, per-system and weighting-sensitivity results accompany the report.

2 Methods

2.1 Tropical-cyclone forecast setting

We study weather-conditioned county-level outage forecasting on the tropical development subset of the public data panel. The subset contains 15 weather systems and 1,633 county–event pairs, including weak systems with little or no reported outage. For each system, a fixed forecast origin separates 72 hours of historical outage observations from a 144-hour forecast window. The target is the public outage fraction constructed by the data pipeline. We retain the pipeline’s observation masks, county sampling probabilities and design weights. The historical information boundary is enforced on all outage-derived inputs.

The weather inputs comprise the available ERA5 and HRRR hazard summaries. Missing weather hours have explicit availability indicators and retain the upstream missing-value convention. Weather availability does not alter the outage scoring mask. Physical county descriptors include terrain, vegetation, soil and coastal-distance variables; six county background variables provide electricity-service and population context. Imputation and standardization are fitted separately within each training set. The 2023 interruption indices are excluded from model inputs. Other static fields and customer denominators retain the public panel’s snapshot vintages.

2.2 Predictors and joint training

Our matched AsymODE reference uses the repository’s two-rate outage-stock model, with damage and recovery networks of widths 32 and 16, respectively. The stock is advanced hourly by

$$p_{i,t} = \text{clip}_{[0,1]} [p_{i,t-1} + u_{i,t}(1 - p_{i,t-1}) - r_{i,t}p_{i,t-1}], \quad (1)$$

initialized at the last observed outage fraction. Both neural predictors use identical weather, historical-outage summary and county-context inputs. The joint predictor additionally introduces the repository’s geography-conditioned response kernel and residual damage/recovery heads, each with 16 hidden units. The residual heads have zero initial output, and the kernel opens gradually; the two predictors therefore have identical predictions at initialization. All parameters are jointly optimized from scratch. This comparison evaluates the combined prediction design, rather than isolating the contribution of geography alone.

Every neural fit uses 900 Adam updates with batches of 128 county–event pairs. Learning rates are 0.003 for damage/kernel parameters and 0.0003 for recovery parameters, with gradient-norm clipping at one. Kernel calibration statistics are refreshed on a fixed training-only sample. The final neural predictions average seeds 0, 1 and 2. These models use reconstructed hazard inputs and minibatch training, so they are distinct from the earlier all-regime HOST experiment.

We also fit gradient-boosted trees to the entire forecast trajectory using lead time as an input. Two residual models use squared-error loss and either ERA5 alone or both weather sources; a direct model uses absolute-error

loss. Residuals are defined relative to exponential-decay persistence, whose single decay parameter is fitted on the relevant training set. Tree settings are fixed at 300 estimators, 15 leaves, learning rate 0.035 and a minimum of 80 samples per leaf. Zero and persistence predictors are reported alongside the learned models.

2.3 Event-held-out evaluation and ensemble selection

We inherit the panel’s five outer event folds without modification. Systems belonging to a linked event group remain together. Inside each outer training set, the four remaining event folds provide inner out-of-fold forecasts. A single vector of nonnegative ensemble weights, summing to one, minimizes the pooled design-weighted inner squared error. These weights remain fixed across all 144 forecast hours. No outer outcome is used to select a weight, checkpoint, seed or candidate configuration. Inner neural fits use seed zero; final refits average three seeds, an approximation whose variance difference is retained as a limitation.

For prediction $\hat{y}_{i,t}$, target $y_{i,t}$, observation indicator $m_{i,t}$ and county–event design weight w_i , we report

$$\text{RMSE} = \sqrt{\frac{\sum_{i,t} w_i m_{i,t} (\hat{y}_{i,t} - y_{i,t})^2}{\sum_{i,t} w_i m_{i,t}}}, \quad (2)$$

$$\text{MAE} = \frac{\sum_{i,t} w_i m_{i,t} |\hat{y}_{i,t} - y_{i,t}|}{\sum_{i,t} w_i m_{i,t}}. \quad (3)$$

Errors are pooled before taking the square root; fold RMSEs are not averaged. We report the full-path metrics, individual point horizons, per-system and exposure-stratum results, and untrimmed-weight and unweighted sensitivities. Paired uncertainty intervals resample the 15 event families 2,000 times with the saved outer predictions fixed. They do not capture repeated model development or refitting uncertainty.

Scope of inference. This is event-held-out development evidence: the same county may occur in other training storms, and prior development results informed the decision to focus on tropical systems. The sealed confirmation tranche remains unread. Future ERA5 and rolling HRRR fields provide conditional hindcast information, not archived forecasts all issued at the outage forecast origin. Static snapshots can also postdate historical storms. The experiment therefore does not establish operational forecasting accuracy or independently confirmed generalization.

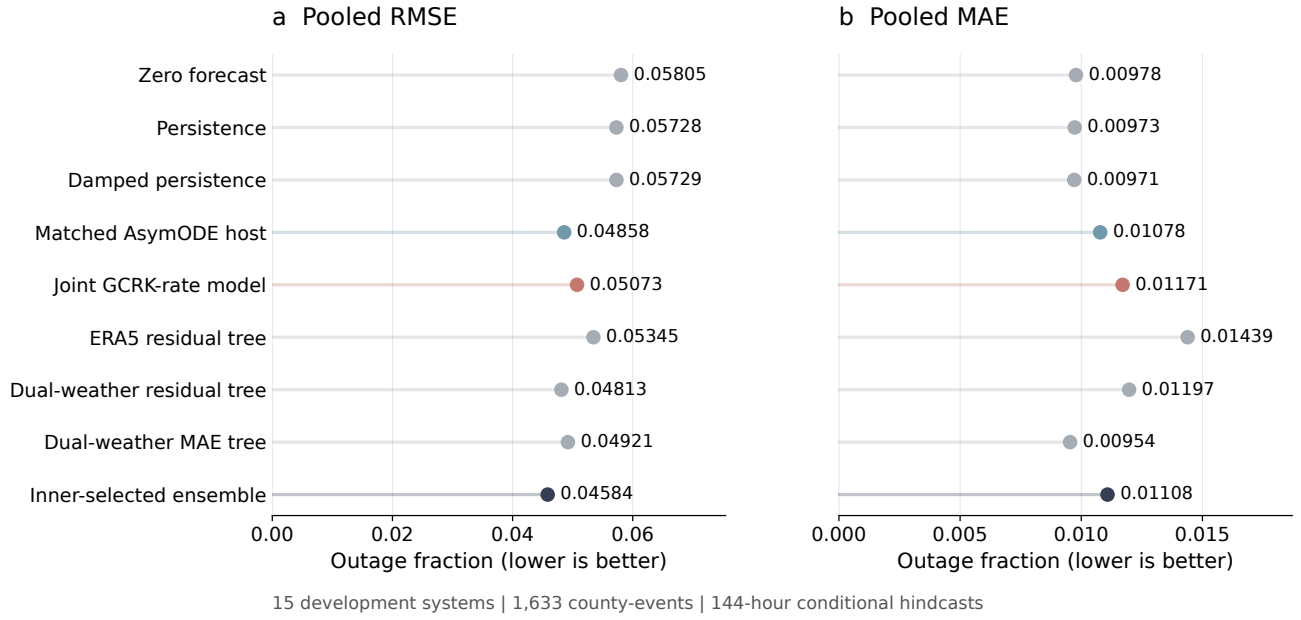


Figure 1: Full-path errors for all fixed candidates. The ensemble weights were selected only from inner event-fold predictions.

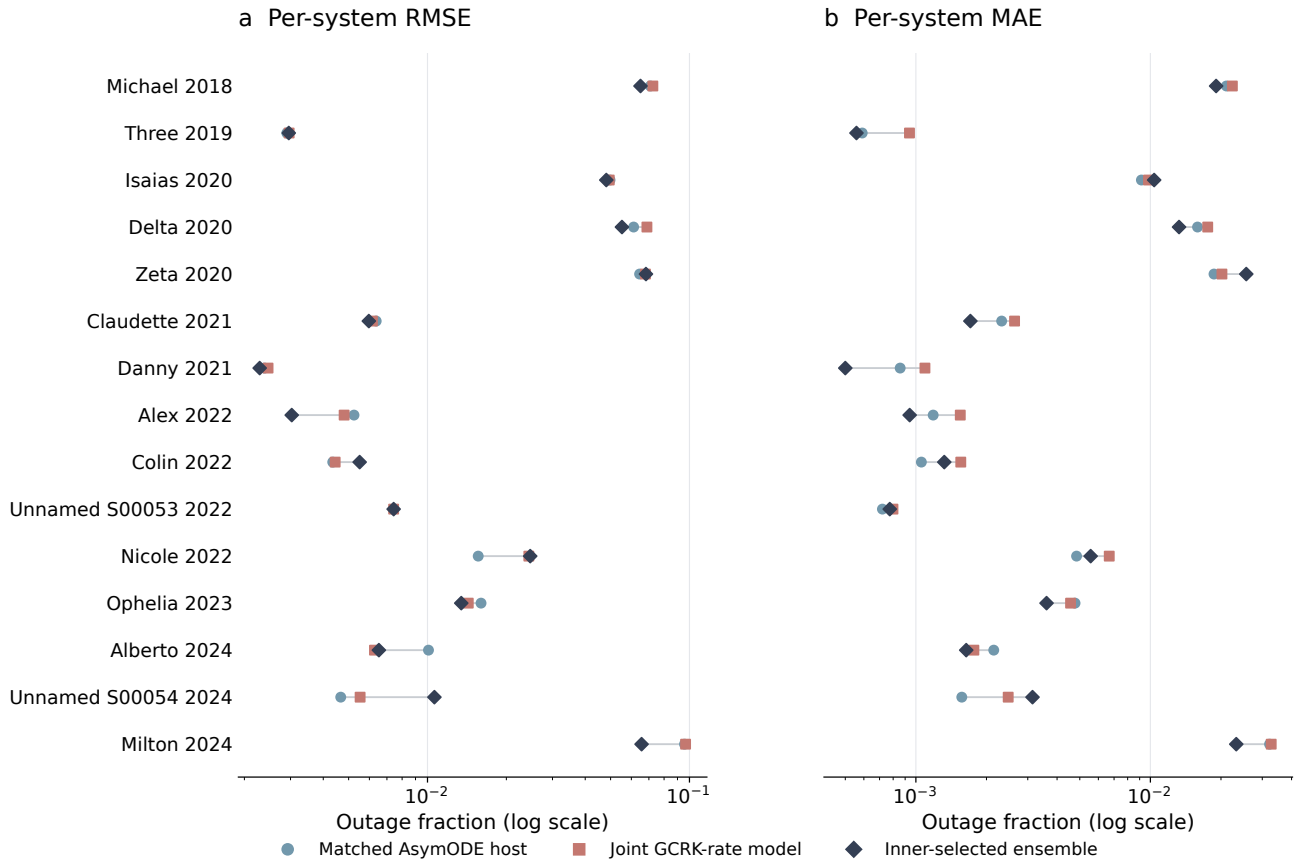


Figure 2: Every tropical development system, including weak systems. Both horizontal axes use logarithmic scales.